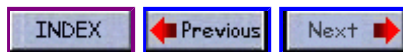


C Programming



DATA TYPES AND CONSTANTS

The four basic data types are

- **INTEGER**

These are whole numbers, both positive and negative. Unsigned integers (positive values only) are supported. In addition, there are short and long integers.

The keyword used to define integers is,

```
int
```

An example of an integer value is 32. An example of declaring an integer variable called **sum** is,

```
int sum;  
sum = 20;
```

- **FLOATING POINT**

These are numbers which contain fractional parts, both positive and negative. The keyword used to define float variables is,

```
float
```

An example of a float value is 34.12. An example of declaring a float variable called **money** is,

```
float money;  
money = 0.12;
```

- **DOUBLE**

These are exponential numbers, both positive and negative. The keyword used to define double variables is,

```
double
```

An example of a double value is 3.0E2. An example of declaring a double variable called **big** is,

```
double big;  
big = 312E+7;
```

- **CHARACTER**

These are single characters. The keyword used to define character variables is,

```
char
```

An example of a character value is the letter **A**. An example of declaring a character variable called **letter** is,

```
char letter;  
letter = 'A';
```

Note the assignment of the character *A* to the variable *letter* is done by enclosing the value in **single quotes**. Remember the golden rule: Single character - Use single quotes.

Sample program illustrating each data type

```
#include < stdio.h >  
  
main()  
{  
    int    sum;  
    float  money;  
    char   letter;  
    double pi;  
  
    sum = 10;           /* assign integer value */  
    money = 2.21;       /* assign float value */  
    letter = 'A';       /* assign character value */  
    pi = 2.01E6;        /* assign a double value */  
  
    printf("value of sum = %d\n", sum );  
    printf("value of money = %f\n", money );  
    printf("value of letter = %c\n", letter );  
    printf("value of pi = %e\n", pi );  
}
```

Sample program output

```
value of sum = 10  
value of money = 2.210000  
value of letter = A  
value of pi = 2.010000e+06
```

